

FORENSIC ACQUISITION REPORT

CASE-001 | 32GB USB Drive Imaging

Examiner: Rahul Kumar

Date: 09-08-2026

1. EXECUTIVE SUMMARY

This report details the forensic acquisition of a 32GB SanDisk Cruzer Blade USB drive. A bit-for-bit physical image was created using FTK Imager 4.5. Cryptographic hash verification (MD5 & SHA-1) confirmed the image's integrity, and a cross-hash validation test on an extracted file further verified logical accuracy.

2. TOOLS USED

- Software: FTK Imager 3.1.2.0 (AccessData)
- OS: Windows 10 Pro (VMware Workstation 16)
- Utility: Windows PowerShell (Hash Verification)

3. PROCEDURE

- The USB drive was attached to the examination VM via USB passthrough.
- FTK Imager was executed with administrative privileges.
- Source type: Physical Drive (selected over Logical to capture unallocated space).
- Output format: Raw (dd) with zero compression and zero fragmentation.
- Destination: F:\Forensics\Case001\USB_32GB_001.001.
- Post-imaging, an automatic verification process was run to compute and compare cryptographic hashes.

4. FINDINGS

- Image File: USB_32GB_001.001 (30.67 GB / 61,341,696 sectors)
- MD5 Checksum: d0a34700e0dd53239f590d1d1b7834bb
- SHA-1 Checksum: 5e1f97a0f653af5dbe09facc1d11e5acafd995a5
- Bad Sectors Detected: None

5. VALIDATION (CROSS-HASH TEST)

A test file ("New Text Document.txt") was extracted from the forensic image. Its MD5 hash was compared to the same file on the original USB drive.

- Original USB Hash: E417C6D3EC20FE1513A32AFE143E52C0
- Image File Hash: E417C6D3EC20FE1513A32AFE143E52C0

Result: MATCH. The logical data within the image is identical to the source.

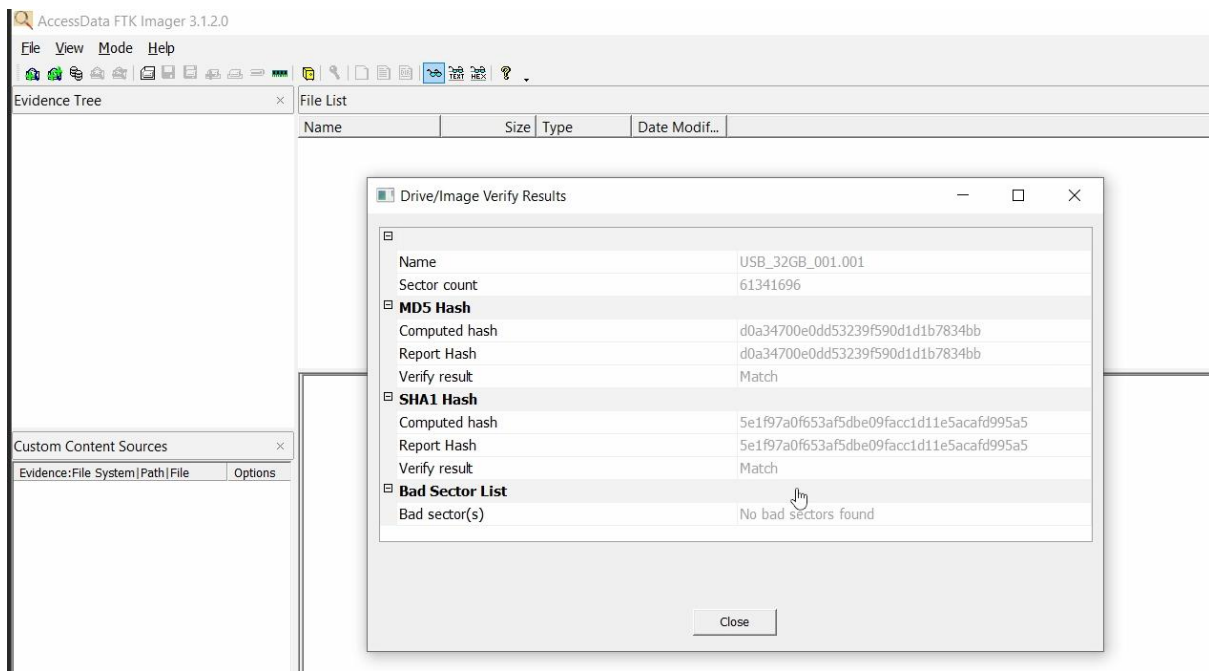
6. EXAMINER'S OPINION

Based on the strict adherence to forensic standards and the successful cryptographic verification, the image "USB_32GB_001.001" is a forensically sound, complete, and unaltered copy of the original media. This evidence is suitable for further analysis and is admissible in a court of law.

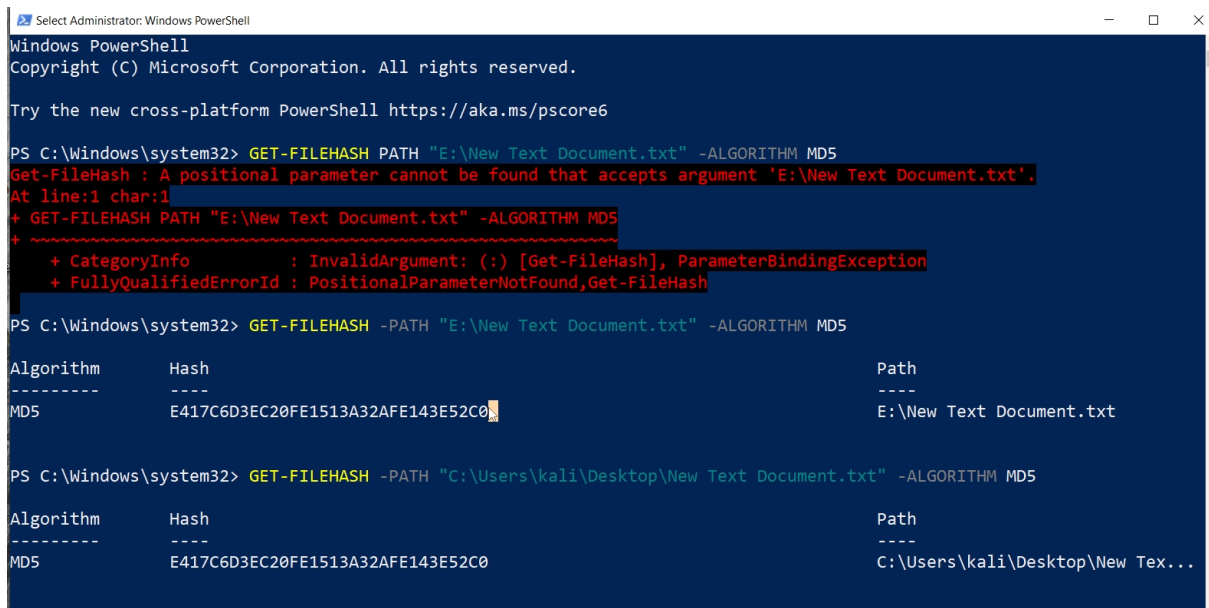
7. ATTACHMENTS

- Chain of Custody Document
- FTK Imager Log File (USB_32GB_001.001.txt)
- Verification Screenshots

Screenshot 1: FTK Verification Pop-up



Screenshot 2: Powershell Cross-Hash Match



Screenshot 3: FTK Log File

USB_32GB_001.001.txt - Notepad

File Edit Format View Help

Created By AccessData® FTK® Imager 3.1.2.0

Case Information:

Acquired using: ADI3.1.2.0

Case Number: CASE-001

Evidence Number: USB-001

Unique description: 32GB USB DRIVE - Forensics Acquisition for Day 1 Project

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Notes: Imaged on windows 10 VM using FTK Imager

Information for F:\Forensics\Case001\USB_32GB_001:

Physical Evidentiary Item (Source) Information:

[Device Info]

Source Type: Physical

[Drive Geometry]

Cylinders: 3,818

Tracks per Cylinder: 255

Sectors per Track: 63

Bytes per Sector: 512

Sector Count: 61,341,696

[Physical Drive Information]

Drive Model: SanDisk Cruzer Blade USB Device

Drive Serial Number: 4C530001141015109093

Drive Interface Type: USB

Removable drive: True

Source data size: 29952 MB

Sector count: 61341696

[Computed Hashes]

MD5 checksum: d0a34700e0dd53239f590d1d1b7834bb

SHA1 checksum: 5e1f97a0f653af5dbe09facc1d11e5acafd995a5

Image Information:

Acquisition started: Sun Aug 09 03:43:21 2026

Acquisition finished: Sun Aug 09 04:32:45 2026

Segment list:

F:\Forensics\Case001\USB_32GB_001.001

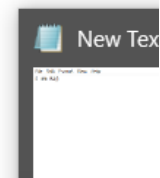
Image Verification Results:

Verification started: Sun Aug 09 04:32:48 2026

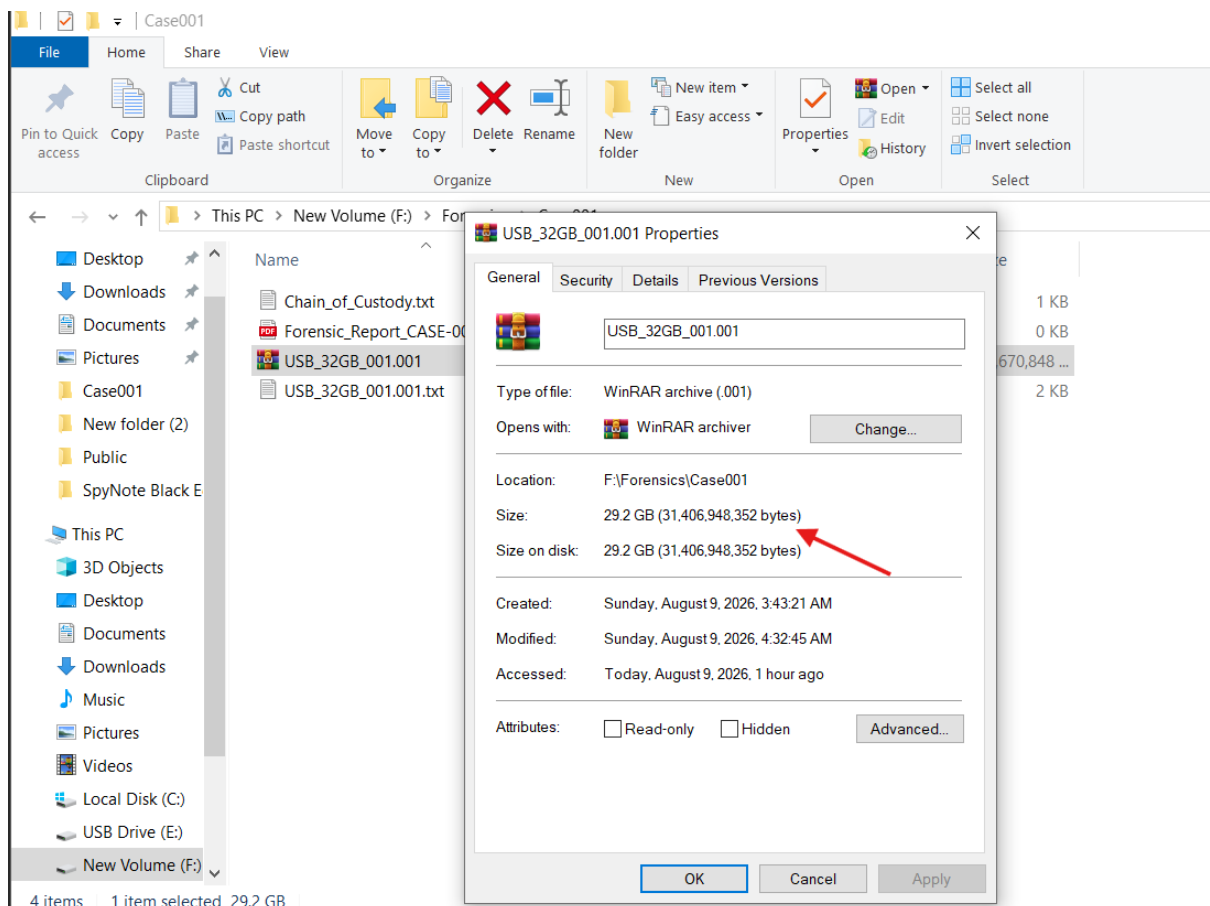
Verification finished: Sun Aug 09 04:41:40 2026

MD5 checksum: d0a34700e0dd53239f590d1d1b7834bb : verified

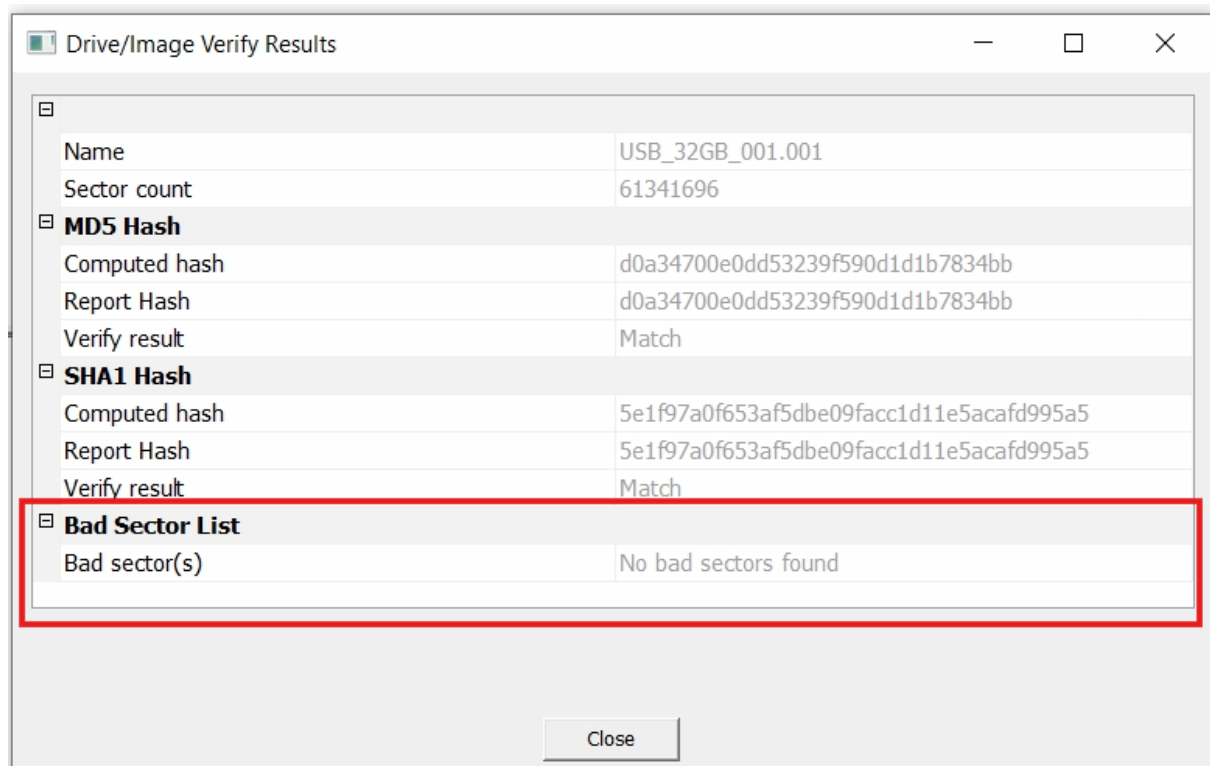
SHA1 checksum: 5e1f97a0f653af5dbe09facc1d11e5acafd995a5 : verified



Screenshot 4: File Explorer – Image Size



Screenshot 5: No Bad Sectors Proof



Examiner Signature: _____
Rahul Kumar

